ABRA — Examination Report

2026-07-25 · Two reviews, adversarial by design. Part I examines the work as a doctoral thesis. Part II examines it as a shipped product. Part III lists every defect with its fix and status.

Nothing here is softened. Every criticism is paired with the evidence that produced it, and where a criticism is *not* supported by evidence it is marked as opinion.

PART I — Thesis examination

Verdict: major revisions. The thesis is not currently defensible, for three reasons that are independent of each other.

The engineering is unusually careful. The measurement discipline, once it exists, exceeds what the published literature in this domain applies to itself. That is genuinely creditable and it is not what fails here.

I.1 The central positive result is arithmetic, not a finding

This is the most serious criticism in this report.

PORY is presented throughout as "the win" — the result that proves the thesis. Its feature vector is:

```
[bias, alive_diff, hp_diff, my_alive, foe_alive, turn/10]
```

That is **the material state of the board**. The claim reduces to: *a logistic regression on how many Pokémon each side has left, and how healthy they are, predicts the winner better than a coin (0.567 vs 0.693).*

Of course it does. If you have three Pokémon and your opponent has one, you are winning. No model is required to know that, and a model that discovers it has discovered arithmetic.

The defence offered is that PORY beats "a material-sign heuristic". That baseline is a strawman: it uses the **sign** of the material difference and discards magnitude, so it cannot distinguish 3-vs-1 from 2-vs-1. Beating it demonstrates that magnitude carries information, which is again not in dispute.

MEASURED 2026-07-25 (`engine/pory_baseline.py`). The answer is worse than "barely".

Identical estimator, identical split by game, identical rows — only the feature set varies:

arm	features	held-out log-loss
B0 coin	0	0.6931
B3 `sign(alive_diff)` — **the baseline the docs used**	1	0.6098
B1 `alive_diff` alone	1	0.6000
B2 `alive_diff + hp_diff`	**2**	**0.5822**
P PORY, full feature set	5	0.5840

PORY LOSES TO A TWO-FEATURE MATERIAL BASELINE. Its three extra features (`my_alive`, `foe_alive`, `turn/10`) make it marginally *worse*, not better. A reviewer can construct B2 in thirty seconds and beat the project's headline model with it.

Caveat, stated because it matters: this reimplementation scores PORY at 0.5840 where the project publishes 0.567, so the shipped fit is better tuned than mine. That does not rescue the finding — every arm above used the **same** estimator, so the comparison is internally valid regardless of how well any of them is tuned. The extra features contribute nothing.

The number to publish is the gain over MATERIAL (+0.02 at best), not the gain over a coin (+0.11). The latter is what currently appears in the white paper, the deck, the summary and the site.

The thesis's own framing makes this worse rather than better. It repeatedly draws the analogy to expected goals in football. xG is a contribution precisely because it is **not** the scoreline — it measures something the observable state does not already tell you. PORY, as specified, is closer to "teams that are ahead tend to win."

I.2 Every negative result is underpowered, and the thesis treats them as findings

The store holds 8,757 games of which 1,124 survive filtering. By the project's own power calculation that detects an edge of no less than ~4.2 accuracy points; a 2-point effect needs ~4,900 games.

Every null the thesis reports sits inside that blind spot: JOLTEON, preview roles, CHOMP-EV, WAR, and the skill ceiling. The thesis states these as properties of the game — "the bring decision sits at the format ceiling", "predicting the winner from sheets is near-impossible". **They are properties of the sample.** The correct statement in every case is "no effect larger than X was detectable at n=Y", and until 2026-07-25 no such statement appeared anywhere.

That the project itself discovered this and built `min_detectable_edge()` is to its credit. That it published five nulls as findings first is the failure, and the published documents still carry the stronger phrasing in places.

A thesis whose central claim is a negative result must own the power analysis as a primary contribution, not add it in the final week.

I.3 The capstone does not exist

The thesis is titled, structured and argued around a decision-support stack culminating in ALAKAZAM. ALAKAZAM is a specification. SLOWKING is, in its own documentation, "a correct chassis, not a bot" — its search is IS-MCTS with documented strategy fusion, a rung below the ReBeL architecture the white paper describes as the target.

So the thesis proposes an in-battle decision engine, spends five documents designing it, and never builds one. What is delivered is a data pipeline, a damage validator, a self-play generator, and a logistic regression on material.

That may be a fine engineering artifact. It is not the thesis that was written.

I.4 Contributions, assessed honestly

Claimed	Assessment
The format is unpredictable from sheets	**Under-evidenced.** Underpowered, and the supporting 55% figure was bot-contaminated
PORY: the live board IS predictable	**Likely trivial.** Features are material state; baseline is a strawman (I.1)
Validated damage engine	**Solid, and genuinely useful.** 31/31 within 2% against two independent oracles
Belief-state search for VGC	**Not novel and not built.** Ihara et al. (2018) compared determinization vs information sets in Pokémon
Bot detection by team invariance	**Genuinely good, and undersold.** A real methodological contribution, demonstrated to change conclusions
Evaluation honesty as a practice	**The strongest contribution, and not framed as one**

The two defensible contributions are the ones the thesis treats as infrastructure: **behavioural bot detection**, and **the measurement discipline**. The demonstration that a headline result (WAR) and a foundational constant (the 55% ceiling) both dissolve under a filter the rest of the field does not apply is worth more than the models.

Reframe the thesis around what actually held. The current framing promises an agent and delivers a methodology; the methodology is the better paper.

I.5 Reproducibility — would fail a replication check

```
requirements.txt   MISSING
.python-version    MISSING
Dockerfile         MISSING
package.json       present
```

There is no pinned Python environment. Numpy was not installed on the author's own machine as of 2026-07-25; the analysis scripts could not run until it was added mid-session. A reviewer attempting replication would not get past setup.

`engine/champions_sim.js` pins a Showdown commit, which is exactly right and shows the author understands the problem. It has simply not been applied to the project's own dependencies.

I.6 Test coverage

```
62 engine files · 9,212 lines · 12 test files
30 engine modules have no test that references them
```

Including `mew.js`, `champions_sim.js`, `set_priors.js`, `analyze.js` — which writes the file the downstream product consumes — and `prior_player.js`, which was found on 2026-07-25 to have been non-constructible under Node for its entire existence.

The tests that exist are unusually good. There are not enough of them, and they cluster on the models rather than the pipeline, which is where every defect this week actually occurred.

PART II — Product examination

Verdict: the site would not survive a technical review, and the reason is that it disagrees with the research it is supposed to present.

II.1 The site cannot state its own dataset size

Three numbers are live simultaneously:

Source	Claim
`web/index.html` line 705	"I count them from **5,199 real Champions games**" — hardcoded
`data/live.js`	`"games": 8757` — the raw store
Measured truth	**8,757 collected, 1,124 usable**

The hardcoded 5,199 is stale by two regenerations. More seriously, **the word "real" is false**: that population is unfiltered, and 75% of the store involves a bot. The site's most prominent credibility claim is its least defensible sentence.

The defensible figure — 1,124 — **appears nowhere on the site.**

II.2 The site presents unfiltered models as filtered

GURU's booth is the one making the "5,199 real games" claim. GURU is one of **28 engines that still read the store raw**. Its matchup matrix therefore includes bot games, in a format where four bot accounts played the same six Pokémon in 1,446 games.

`data/guru.js` and `data/xatu.js` were last generated **2026-07-23** — before the store was repaired, before the quality filter was applied to five engines, and before the metagame model was corrected. The site is serving two-day-old numbers computed on a store that has since been rebuilt.

II.3 The third damage engine is still live and still untested

`web/index.html` contains 7 references to `dmgRange` / `mcWinProb` — the embedded copy of the rollout engine that `ARCHITECTURE.md` fault 1.1 identified as the project's largest structural violation. The contract test covers the CHOMP and MEDICHAM implementations. **It does not cover the site's copy.**

So the artifact a user actually interacts with runs a rules engine that nothing verifies, which already disagreed with its sibling by 30% on a mega's Special Attack once before.

II.4 `app/` and `web/` are synchronised by hand

The documented procedure is `cp web/index.html app/index.html`. They are currently identical, and nothing enforces it. A single forgotten copy ships a site that differs from the reviewed source, with no failing check. This is S1, in the most user-visible place in the project.

II.5 The new work is not wired to the product

Shipped 2026-07-25 and **not reachable from the site**: the two-view metagame model, MEW's self-play corpus, the Smogon priors, the corrected spreads. `web/mew.html` exists as a standalone page with no link from the main site.

A CTO's summary would be: *the research improved substantially this week and the product did not change at all.*

II.6 What works

Stated plainly because the criticism above is heavy. The buildless single-file architecture is a legitimate and well-executed choice for this project. PORY's live win-probability readout is genuinely good product design. KADABRA working offline from a bundled corpus is thoughtful. `mew.html` states what MEW *cannot* do as prominently as what it can, which is rarer than it should be.

PART III — Defects and dispositions

#	Defect	Severity	Status
1	PORY loses to a two-feature material baseline	**Critical**	**MEASURED and confirmed** — see I.1. The headline result does not survive
2	Site hardcoded "5,199 real Champions games" — stale AND unfiltered	**Critical**	**FIXED** — now states 8,757 collected / 1,124 usable, and that GURU is unfiltered
3	Site never showed the usable count	High	**FIXED**
4	Engines bypassing the quality filter	High	**PARTIAL** — GURU, XATU, archetypes wired via new engine/store.py (single reader). GURU now shows 0 decisive matchups. ~25 remain
5	Third damage engine uncovered by contract test	High	**FIXED** — tests/test-site-engine.js extracts MEDI2 from the site and compares it to the canonical engine. 0.00% divergence across 8 scenarios
6	`guru.js` had NO GENERATOR and was two days stale	High	**FIXED** — build/build_guru_js.js written; both regenerated on filtered data
7	No pinned Python environment	High	**FIXED** — requirements.txt + .python-version
8	`app/` and `web/` synchronised by hand	Medium	**FIXED** — tests/test-site-sync.js
9	30 engine modules have no test	Medium	Open
10	New work not linked from the site	Medium	**FIXED** — MEW panel added, mew.html copied to app/
11	Nulls stated as findings in published prose	Medium	Partially fixed
12	S11 one-publisher violated by the auto-commit	Low	Open, documented

The three that matter most, in order: defect 1, because it may remove the thesis's only positive result; defect 4, because every published meta number depends on it; defect 5, because it is the artifact users actually touch.